Mines Paris PSL - Etilab

Pooling Contracts between Venture Capital Institutions

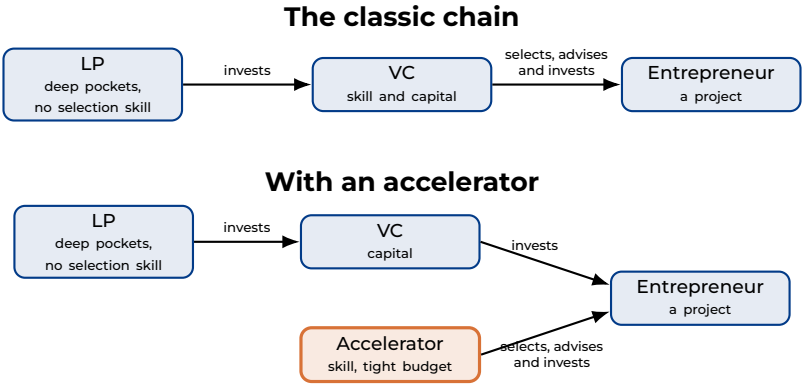
Stephen Dossou
Joint with Pierre Fleckinger

November 5, 2026

Outline

1. Introduction
2. Literature Review
3. Contribution
4. Model
5. Policy
6. Takeaways

Who does what in venture capital



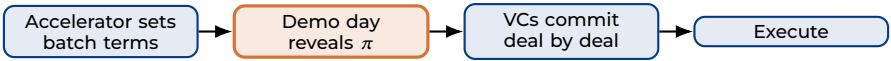
Contracts between VC institutions

- VCs and limited partners
  - LPs commit before seeing any target company, in 93% of deals *(Lerner et al., 2022)*. Commitment is ex ante.
  - The VC invests 1–5% of the total cost, the same for every project.
  - The VC receives 20% of the return plus 2% of committed capital, the same for every project.
- Accelerators and follow-on VCs
  - A demo day reveals the batch to follow-on VCs before they invest. Commitment is ex post.
  - Terms are posted in advance and identical within a batch:

	Fixed amount, fixed share	Contingent tranche
Y Combinator	$125K for 7%	$375K uncapped MFN safe, 2022–
Techstars	$20K for 6%, 5% since 2025	$100K note capped at $3M, then MFN
500 Global	$150K for 6%	—
Entrepreneur First	$125K for 8%, London	$125K uncapped MFN safe
Antler	£125K for 8.5%, UK	£85K convertible note

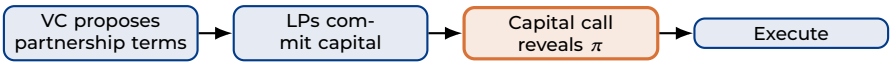
Two relationships, two timings

Accelerators and follow-on VCs



Investors learn π before they commit: ex-post commitment.

VCs and limited partners



Investors commit before they learn π: ex-ante commitment.

Literature Review

- Performance of contracts between VCs and LPs
  - Lerner and Nanda (2020), Metrick and Yasuda (2010): management fees push funds to grow at the expense of performance.
  - Axelson et al. (2009), Fang et al. (2015): blind pool contracts reduce VCs' incentives to overstate project values.
  - Gourier et al. (2024), Lerner and Nanda (2020): LPs' demand for VC is high and concentrated.
- Emergence and performance of the accelerator model
  - Ewens, Nanda and Rhodes-Kropf (2018): cheaper experiments move early-stage investors to smaller initial stakes in more startups.
  - Gonzalez-Uribe and Leatherbee (2017): accelerators increase VC fundraising and scale.
  - Kim and Wagman (2014): an accelerator picks a smaller portfolio than the efficient one and may disclose only its good signals.
  - Zarei, Rasti-Barzoki and Moon (2022): screening is the accelerator's service that matters most.
- Theory of contracts
  - Myerson (1983): an informed principal can propose one mechanism whatever the projects it holds.
  - Maskin and Tirole (1992), Holmstrom and Myerson (1983), Guesnerie and Laffont (1984), Kamien and Schwartz (1991).

This paper

We study contracts between venture capital institutions in an informed-principal model with competitive investors.

Results

- Under both regimes, pooling contracts are strategically optimal for the agency.
- Under ex-post commitment:
  - the unique optimal contract is pooling,
  - the agency's budget is always saturated,
  - some good projects always remain unfunded.

The agents

Entrepreneurs

A project needs a sunk cost $I \in (0,1)$, yields an outcome in $\{0,1\}$ with success probability π. Density f, c.d.f. F. No effort cost: the entrepreneur needs no equity.

The agency

- Budget B_a
- Knows π for each project
- Profit-maximizing

Investors

- Budget B_i
- Know only the distribution of π
- Competitive (price-taking)

Allocation $\mu = (\sigma, \rho)$: the agency contributes $\sigma(\pi)$ to the cost I and keeps a share $\rho(\pi)$ of the outcome. The investor pays the rest of the cost and receives the rest of the outcome.

Payoffs and assumptions

Payoffs for each project π:

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \qquad U_i(\pi) = \big(1 - \rho(\pi)\big)\pi - \big(I - \sigma(\pi)\big)$$

Budget constraints: $B_a \geq \int \sigma(\pi) f(\pi)\, d\pi$ and $B_i \geq \int (I - \sigma(\pi)) f(\pi)\, d\pi$

Assumptions

(A1) $\mathbb{E}(\pi) - I < 0 < 1 - I$, and f is log-concave. The average project is not worth financing, the best one is.

(A2) $B_a < I(1 - F(I)) \leq B_a + B_i$. The agency cannot finance all good projects alone, but the agency and the investors together can.

(A3) The agency is profit-maximizing.

Game 1 (Benchmark)

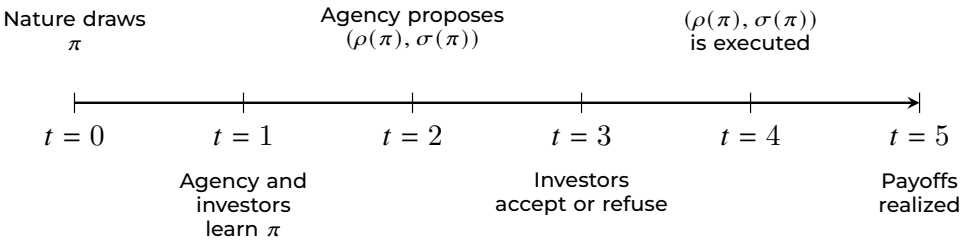


Figure: Game 1 – Benchmark

Game 1 (Benchmark)

Proposition 1

Under budget and investors participation constraints:

The agency maximizes profit	$\Leftrightarrow$	Investor utility $U_i(\pi) = 0$ for each π
		The agency finances all good projects

Investors participation constraint: $U_i(\pi) \geq 0$ for each π

Game 1 (Benchmark)

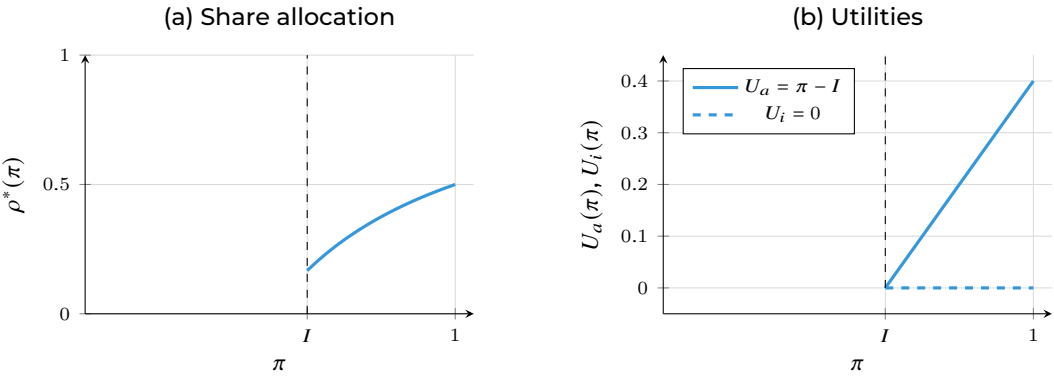


Figure: Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in π. (b) The agency captures the full surplus $\pi - I$; the investor earns zero rent.

Game 2 (Ex-post commitment)

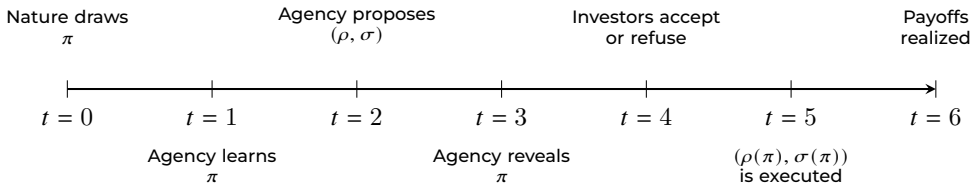


Figure: Game 2 – Ex-post commitment

Game 2 (Ex-post commitment)

Proposition 2

Under budget and investors participation constraints:

The agency maximizes profit $\Leftrightarrow$

- The allocation is pooling
- The agency saturates its budget
- $U_i(\pi_{\min}) = 0$ for the less good financed project
- The profit-maximization FOC is satisfied

Investors participation constraints:

- $U_i(\pi) \geq 0$ for each π
- $U_a(\pi) \geq \rho(\hat{\pi})\pi - \sigma(\hat{\pi})$ for each $(\pi, \hat{\pi})$

Game 2 (Ex-post commitment)

Remark

$\pi_{\min} > I$, so some good projects always remain unfunded.

A contract for each project

Four projects, $I = 1/2$, $B_a = 0.4$. Tailored terms, every investor exactly at zero.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	$(0.1,\ 0.05)$	$(0.1,\ 0.05)$	$(0.4,\ 0.05)$	$(0.6,\ 0.1)$	
Agency U_a	0	0	0.25	0.50	$\Sigma = 0.75$
Investor U_i	0	0	0	0	$\Sigma = 0.00$

The agency takes the whole surplus, 0.75, and spends 0.25 of its budget.

The agency is the one who says which project is which

It reports a $\pi = 1/2$ project as $\pi = 1$: $0.6 \cdot \frac{1}{2} - 0.1 = 0.20 > 0$.

reported as $\pi = 1$

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	$(0.1,\ 0.05)$	$(0.6,\ 0.1)$	$(0.4,\ 0.05)$	$(0.6,\ 0.1)$	
Agency U_a	0	0.20	0.25	0.50	$\Sigma = 0.95$
Investor U_i	0	-0.20	0	0	$\Sigma = -0.20$

Every project below $\pi = 1$ wants that contract, and the constraint $U_i(\pi) \geq 0$ is written project by project.

One contract for all, and a saturated budget

The best single contract is $(0.6,\ 0.2)$: it funds $\pi \geq 3/4$ and spends $2 \times 0.2 = B_a$.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	not funded	not funded	$(0.6,\ 0.2)$	$(0.6,\ 0.2)$	
Agency U_a	—	—	0.25	0.40	$\Sigma = 0.65$
Investor U_i	—	—	0	0.10	$\Sigma = 0.10$

The contract pools, saturates the budget and sets $U_i(\pi_{\min}) = 0$, with $\pi_{\min} = 3/4 > I = 1/2$.

Game 2 (Ex-post commitment)

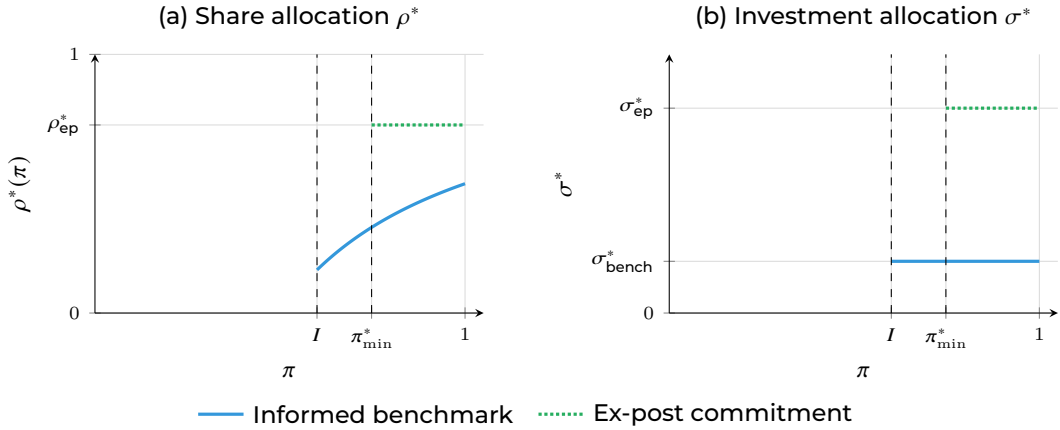


Figure: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0,1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$.

Game 2 (Ex-post commitment)

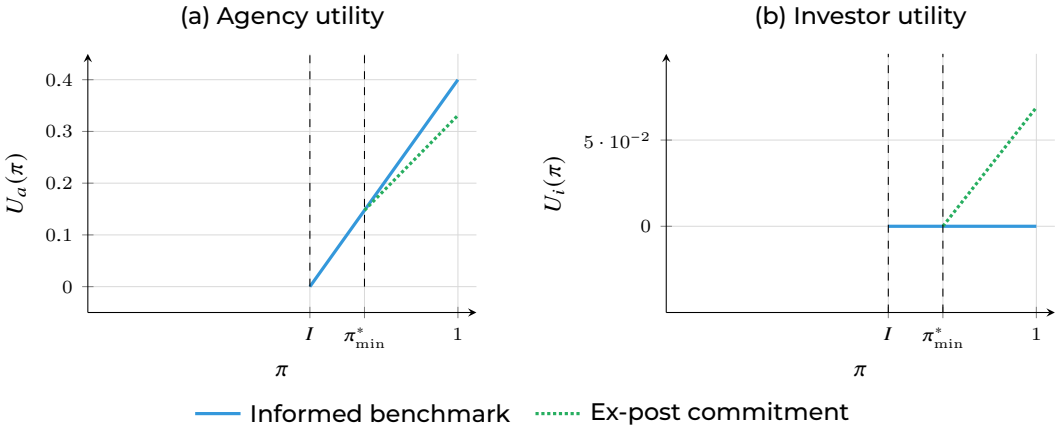


Figure: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters.

Game 3 (Ex-ante commitment)

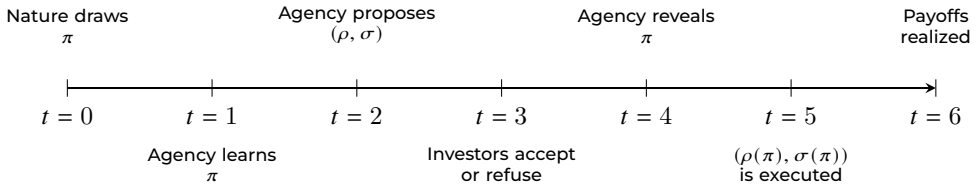


Figure: Game 3 – Ex-ante commitment

Game 3 (Ex-ante commitment)

Proposition 3

Under budget and investors participation constraints:

The agency maximizes profit $\Longleftarrow$

- The allocation is pooling
- The agency does not saturate its budget
- Investor average utility $\int U_i(\pi) f(\pi) d\pi = 0$
- The agency finances all good projects

Investors participation constraint: $\int U_i(\pi) f(\pi) d\pi \geq 0$

Game 3 (Ex-ante commitment)

Remark

The reverse is not true.

Game 3 (Ex-ante commitment)

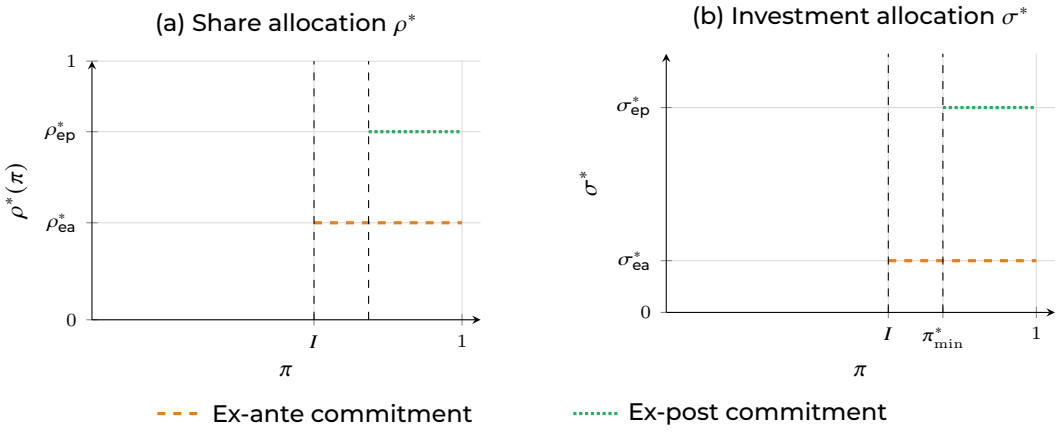


Figure: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as in Game 2.

Game 3 (Ex-ante commitment)

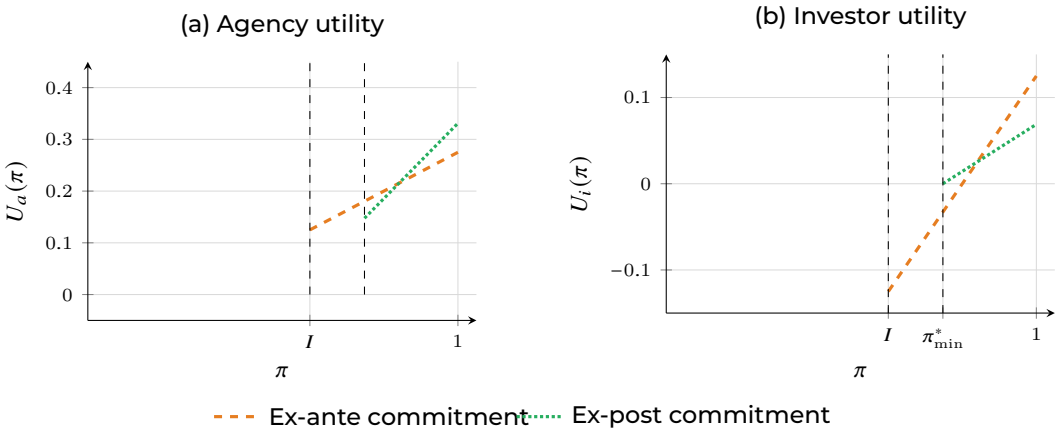
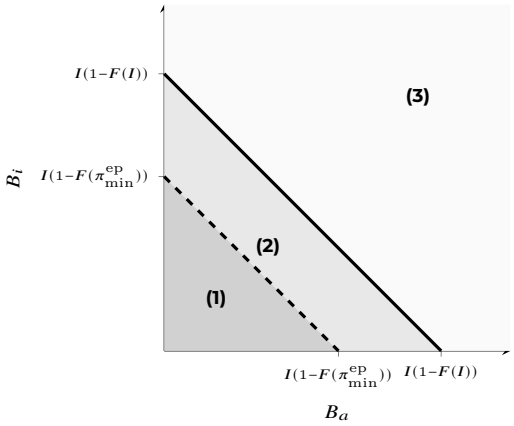


Figure: Utilities under the two commitment regimes. Same parameters as in Game 2.

Cash-poor vs. cash-rich economies

The friction is slack only where capital itself is short.



(1) Severely cash-poor: both regimes coincide
(2) Moderately cash-poor: ex-post excludes more
(3) Cash-rich: ex-post excludes $[I, \pi^{\text{ep}}_{\min})$

Takeaways

Main result

- Ex-ante: pooling is optimal, but not uniquely.
- Ex-post: pooling is unique, budget-saturating, inefficient.

- Pooling contracts are optimal.
- Accelerators as early-stage financiers induce an inefficiency.
- Public capital can efficiently intervene as LP.

Thank you

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Venture Capital Institutions

Stephen Dossou

Joint with Pierre Fleckinger
Mines Paris PSL – Etilab

stephen.dossou@minesparis.psl.eu

stephendossou@gmail.com

eti|lab

MINES PARIS | PSL